

CS223 Computer Architecture & Organization

Cache Memory Design Parameters



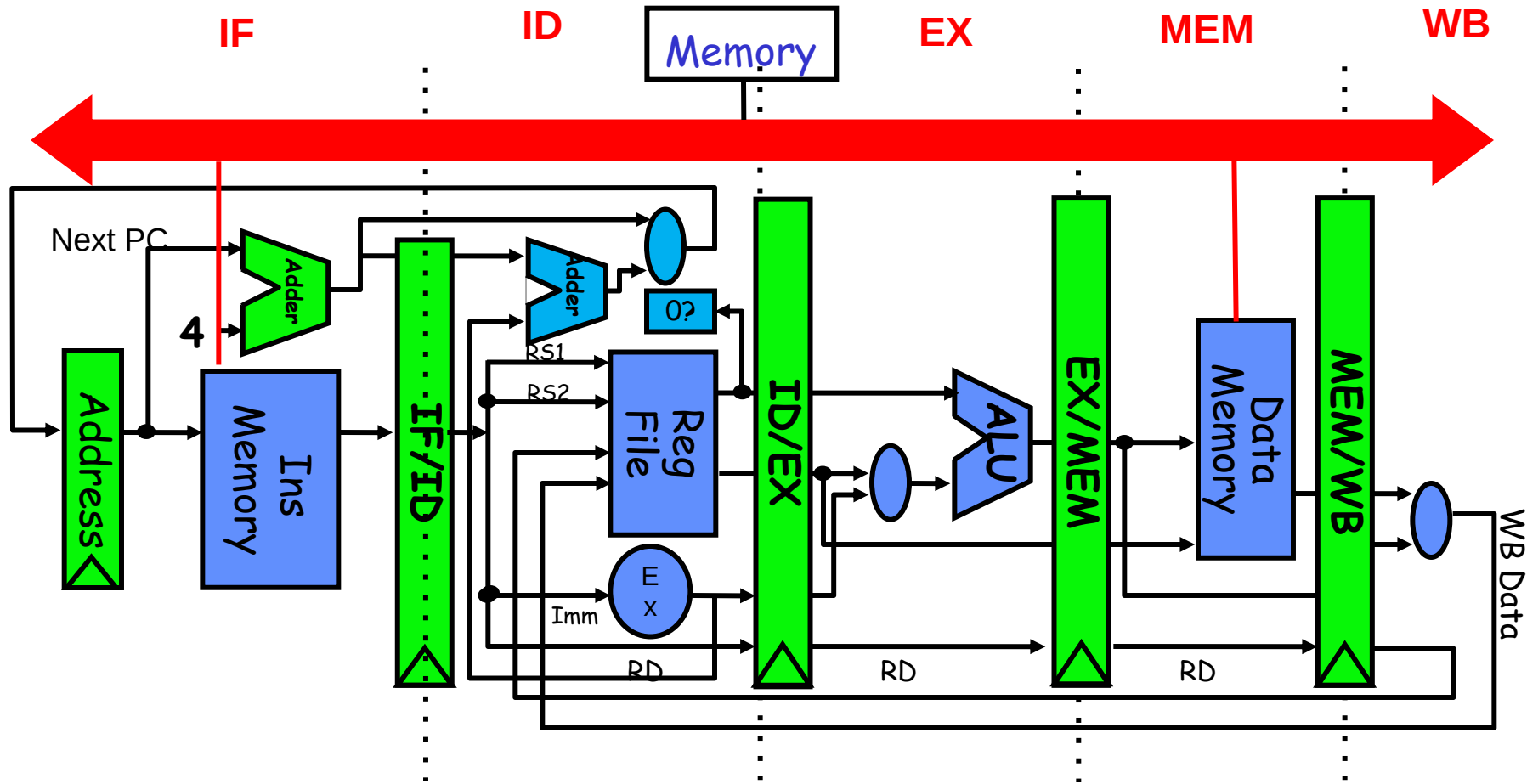
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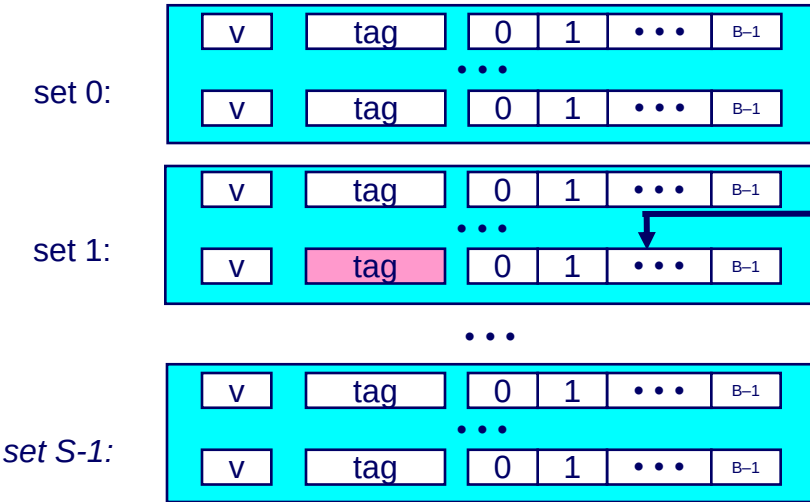
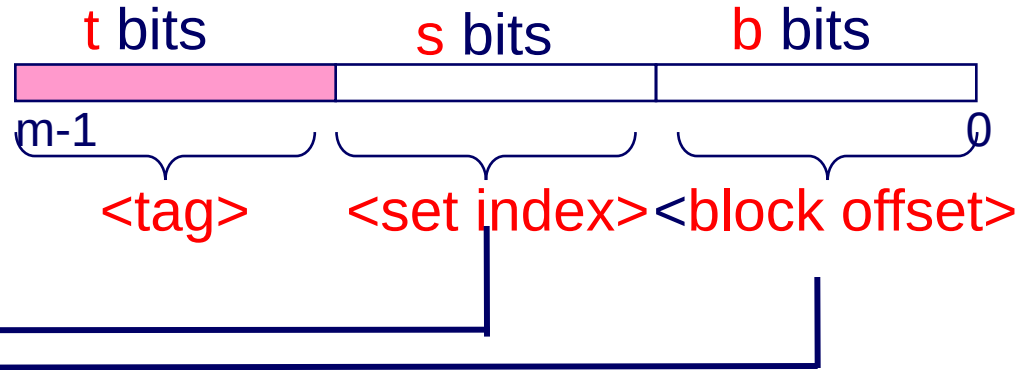
Indian Institute of Technology Guwahati

Relationship of Caches and Pipeline



Addressing Caches

CPU wants → Address A:



- ❖ The word at **address A** is in the cache if the tag bits in one of the **<valid>** lines in set **<set index>** match **<tag>**
- ❖ The word contents begin at offset **<block offset>** bytes from the beginning of the block

Four cache memory design choices

- ❖ Where can a block be placed in the cache?
 - **Block Placement**
- ❖ How is a block found if it is in the upper level?
 - **Block Identification**
- ❖ Which block should be replaced on a miss?
 - **Block Replacement**
- ❖ What happens on a write?
 - **Write Strategy**

Block Placement

Block Number

0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 2 2 2 2 2 2 2 2 2 2 2 3 3

Memory

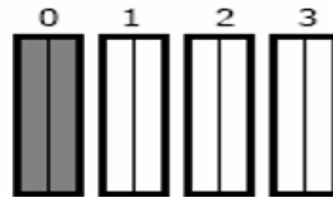


Set Number

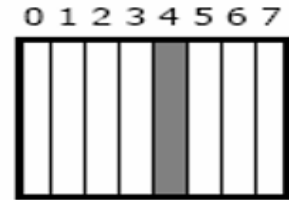
Cache



Fully
Associative



(2-way) Set
Associative



Direct
Mapped

block 12
can be placed

anywhere

anywhere in
set 0
($12 \bmod 4$)

only into
block 4
($12 \bmod 8$)

Cache Mapping / Block Placement

❖ Direct mapped

- ❖ Block can be placed in only one location
- ❖ $(\text{Block Number}) \bmod (\text{Number of blocks in cache})$

❖ Set associative

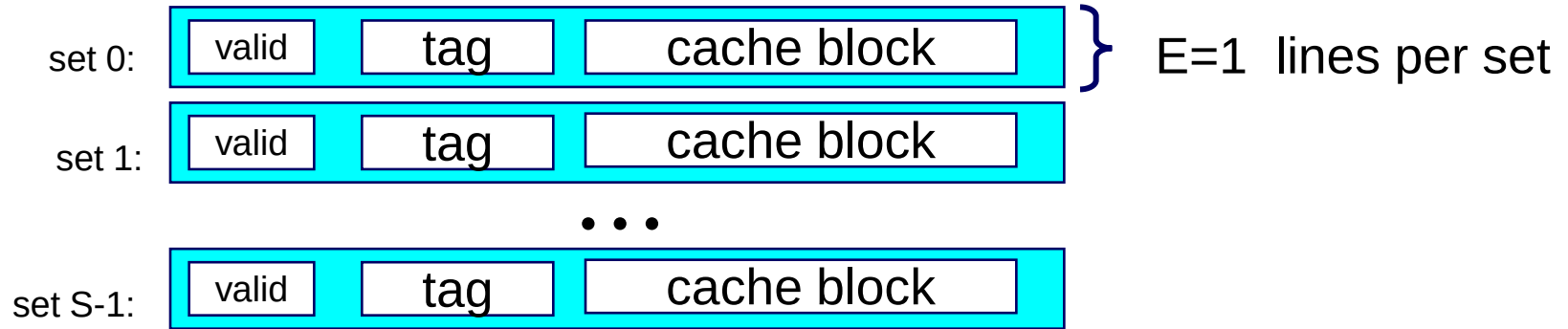
- ❖ Block can be placed in one among a list of locations
- ❖ $(\text{Block Number}) \bmod (\text{Number of sets})$

❖ Fully associative

- ❖ Block can be placed anywhere

Direct-Mapped Cache

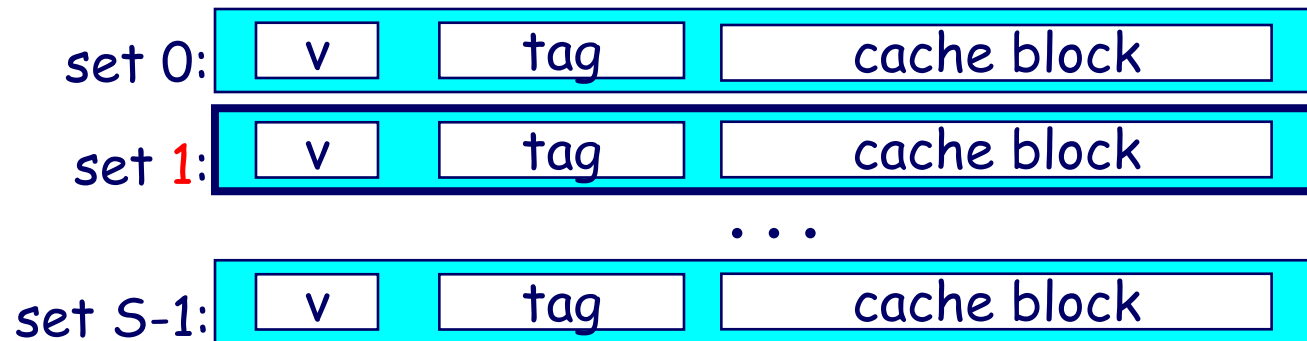
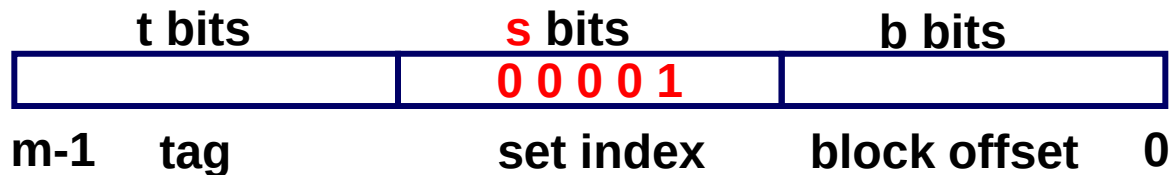
- ❖ Simplest kind of cache, easy to build
- ❖ Only 1 tag compare required per access
- ❖ Characterized by exactly one line per set.



Cache size: $C = B \times S$ data bytes

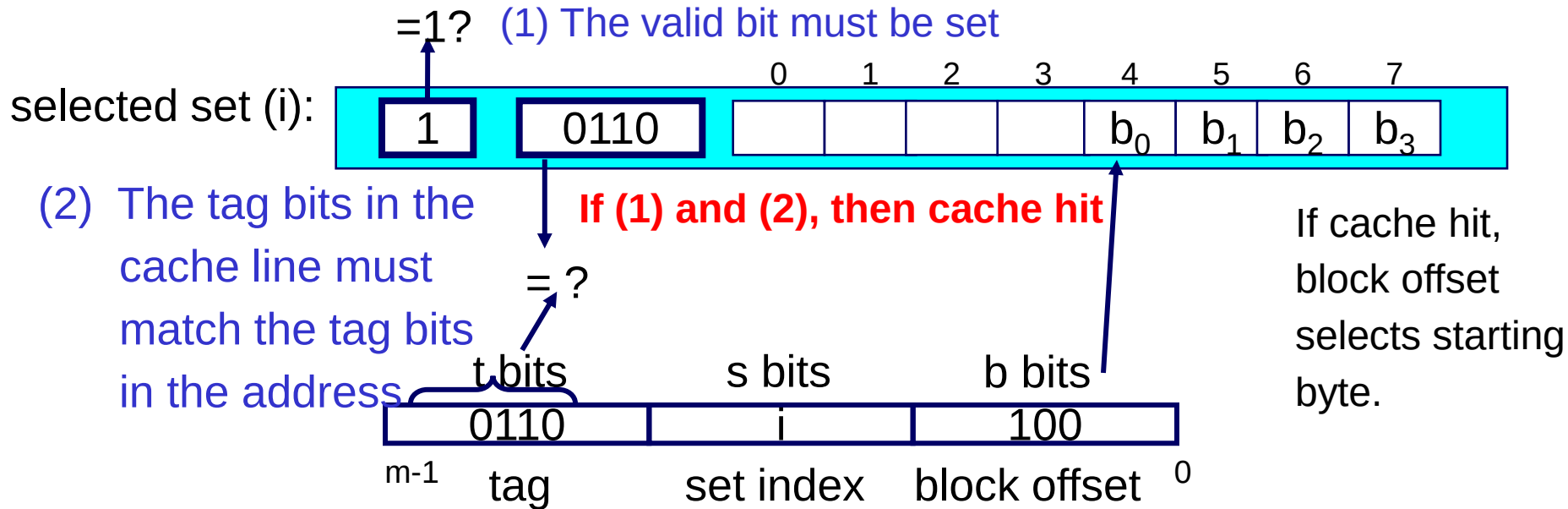
Accessing Direct-Mapped Caches

- ❖ Set selection is done by the set index bits

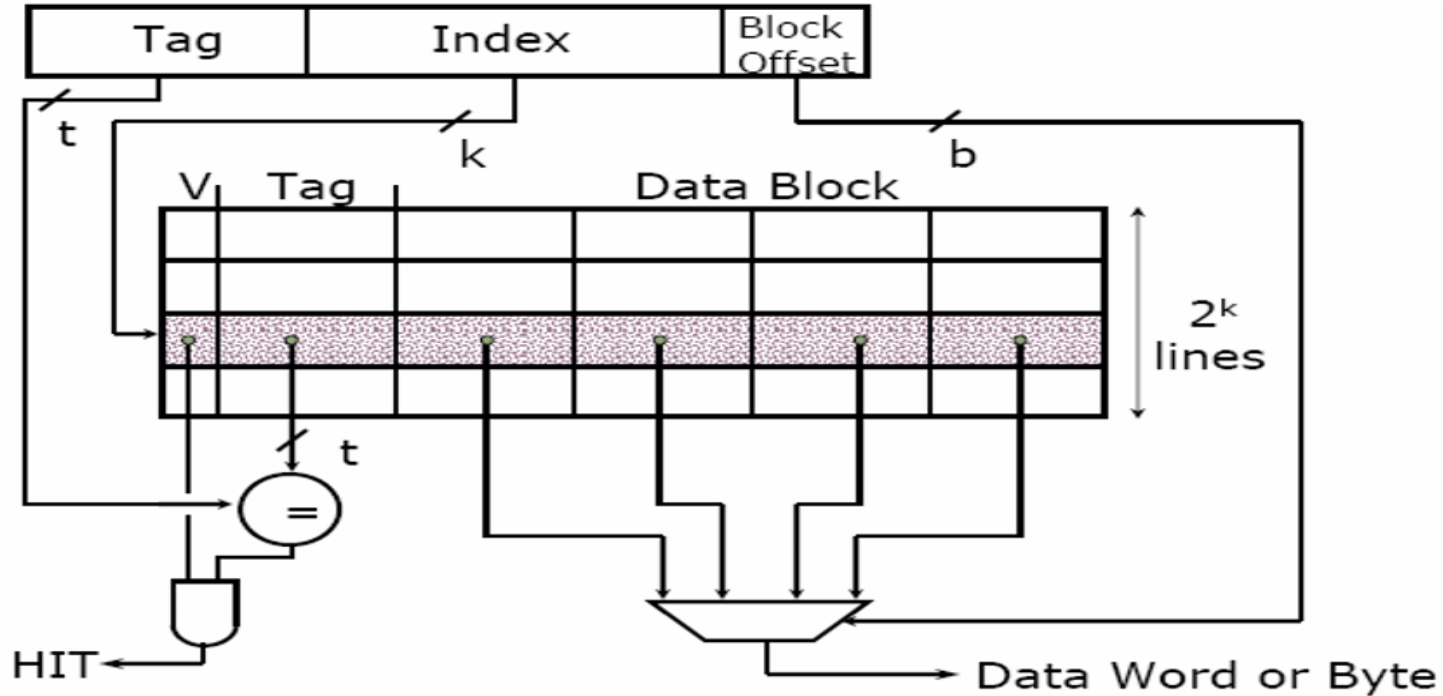


Accessing Direct-Mapped Caches

- ❖ **Block matching:** Find a valid block in the selected set with a matching tag
- ❖ **Word selection:** Then extract the word

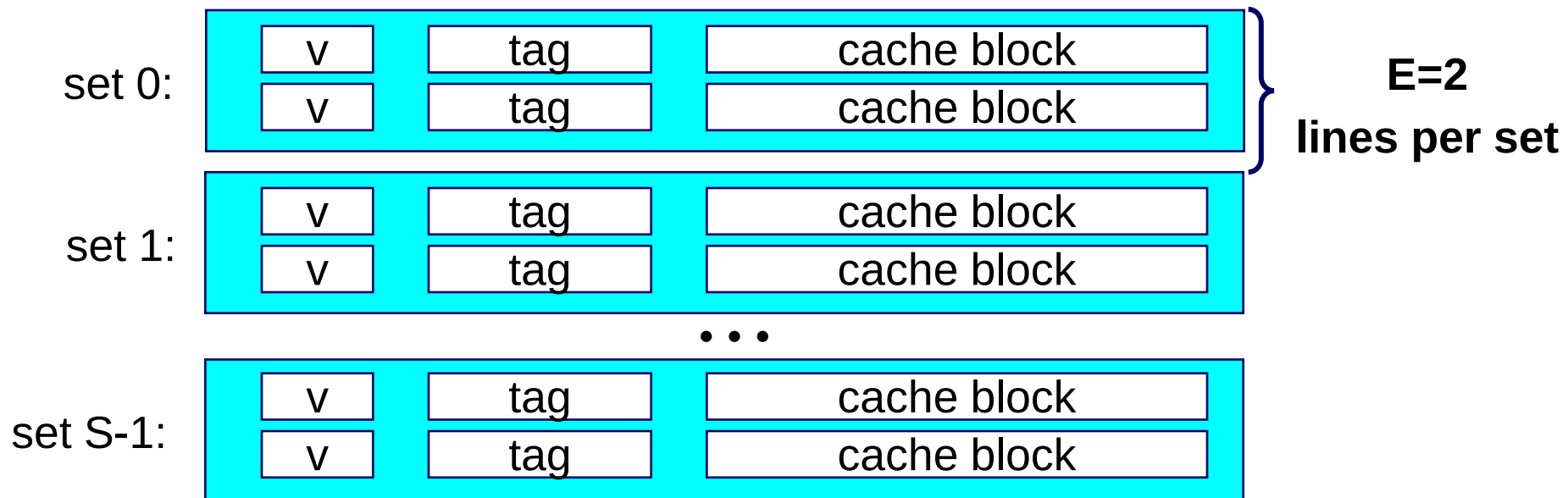


Block Identification – Direct mapped



Set Associative Cache

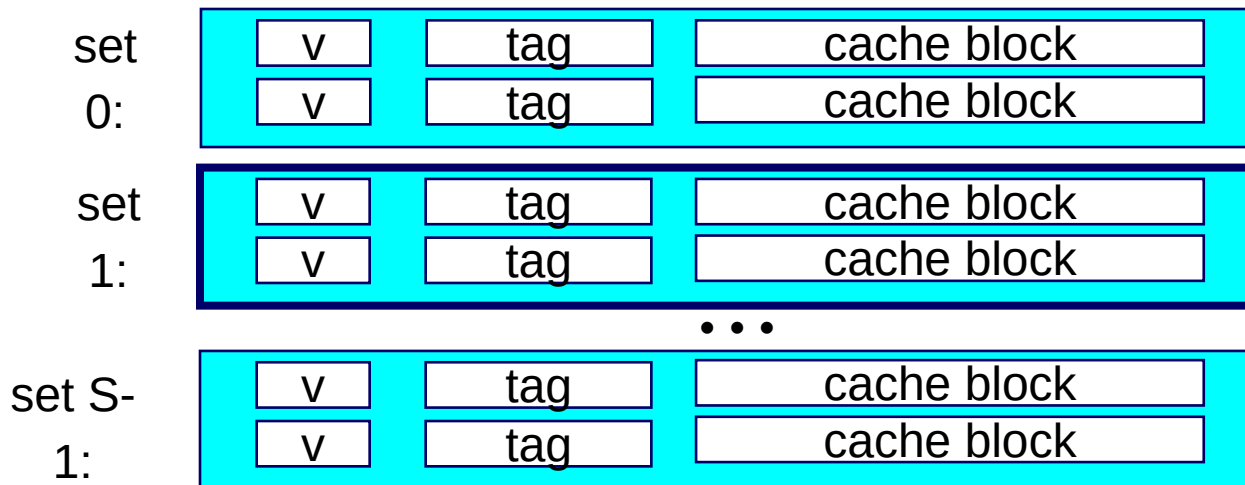
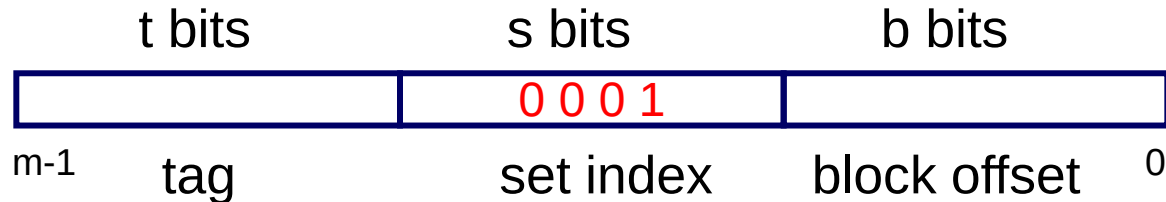
- ❖ Characterized by more than one line per set



2-way associative cache

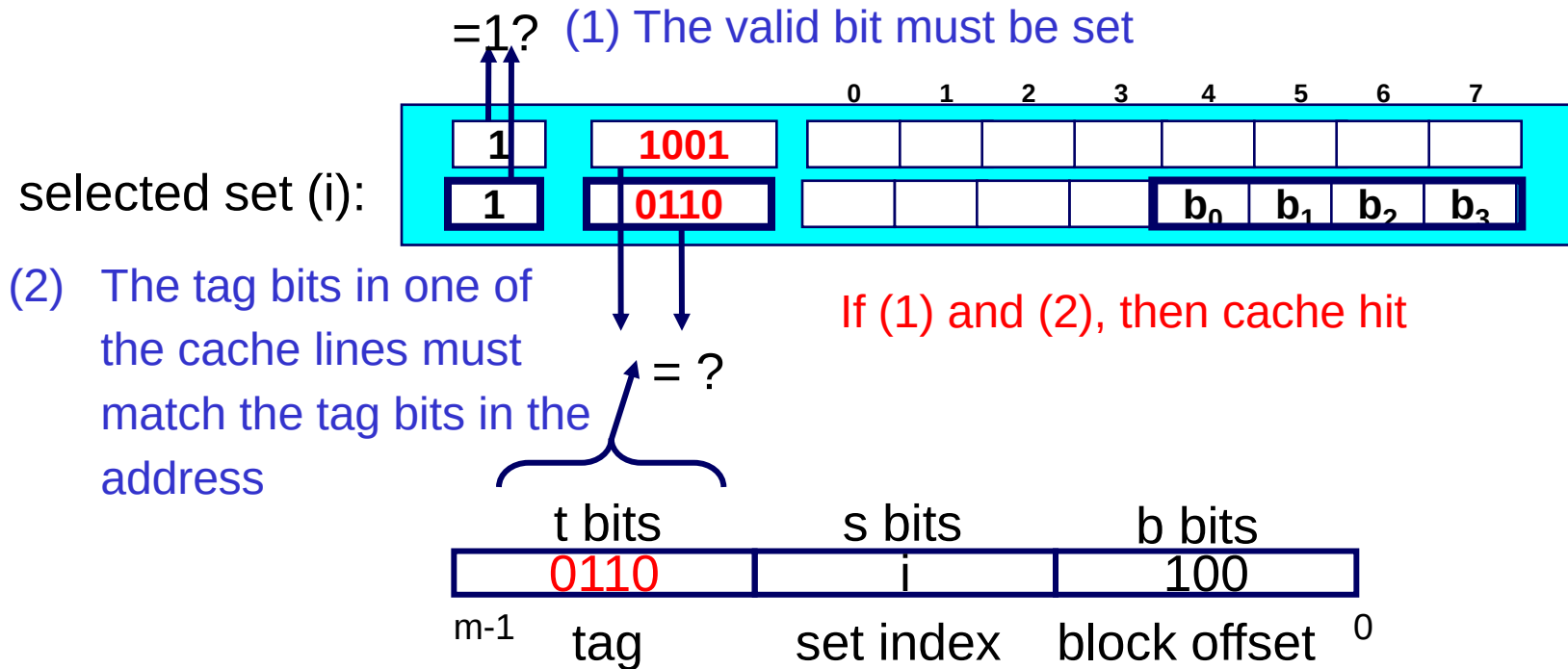
Accessing Set Associative Caches

- ❖ Set selection is identical to direct-mapped cache



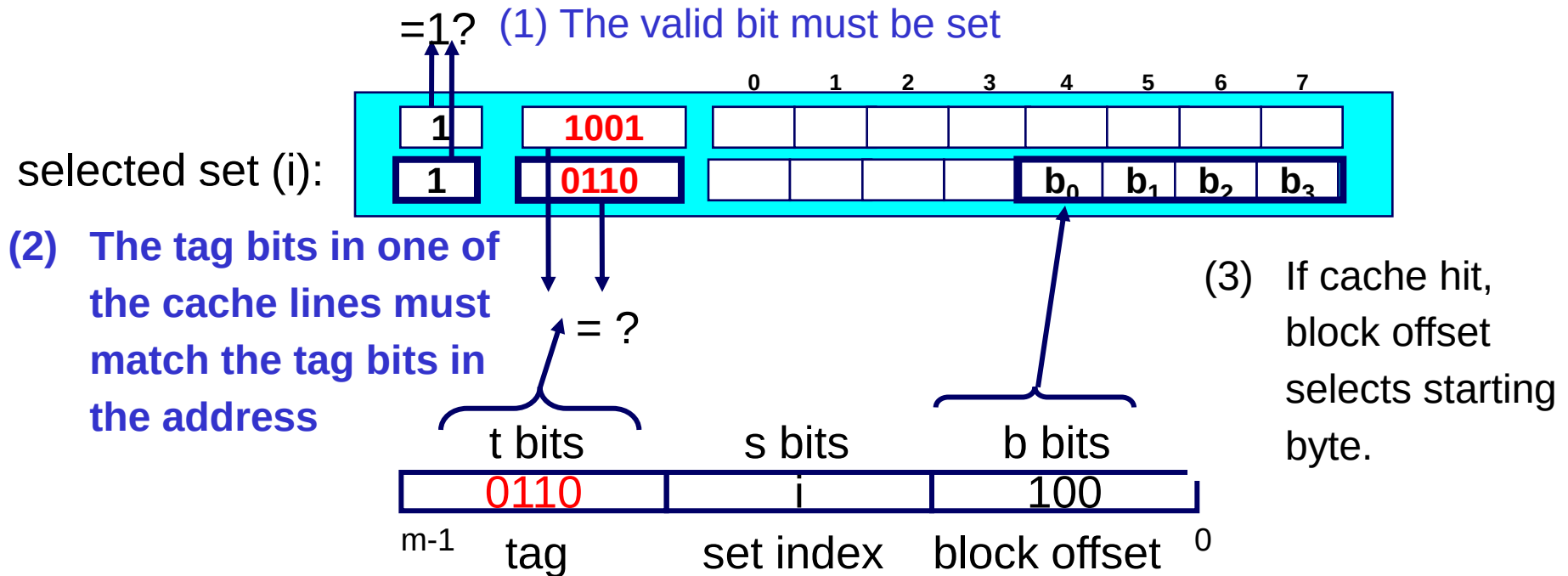
Accessing Set Associative Caches

- ❖ Block matching is done by comparing the tag in each valid line in the selected set.

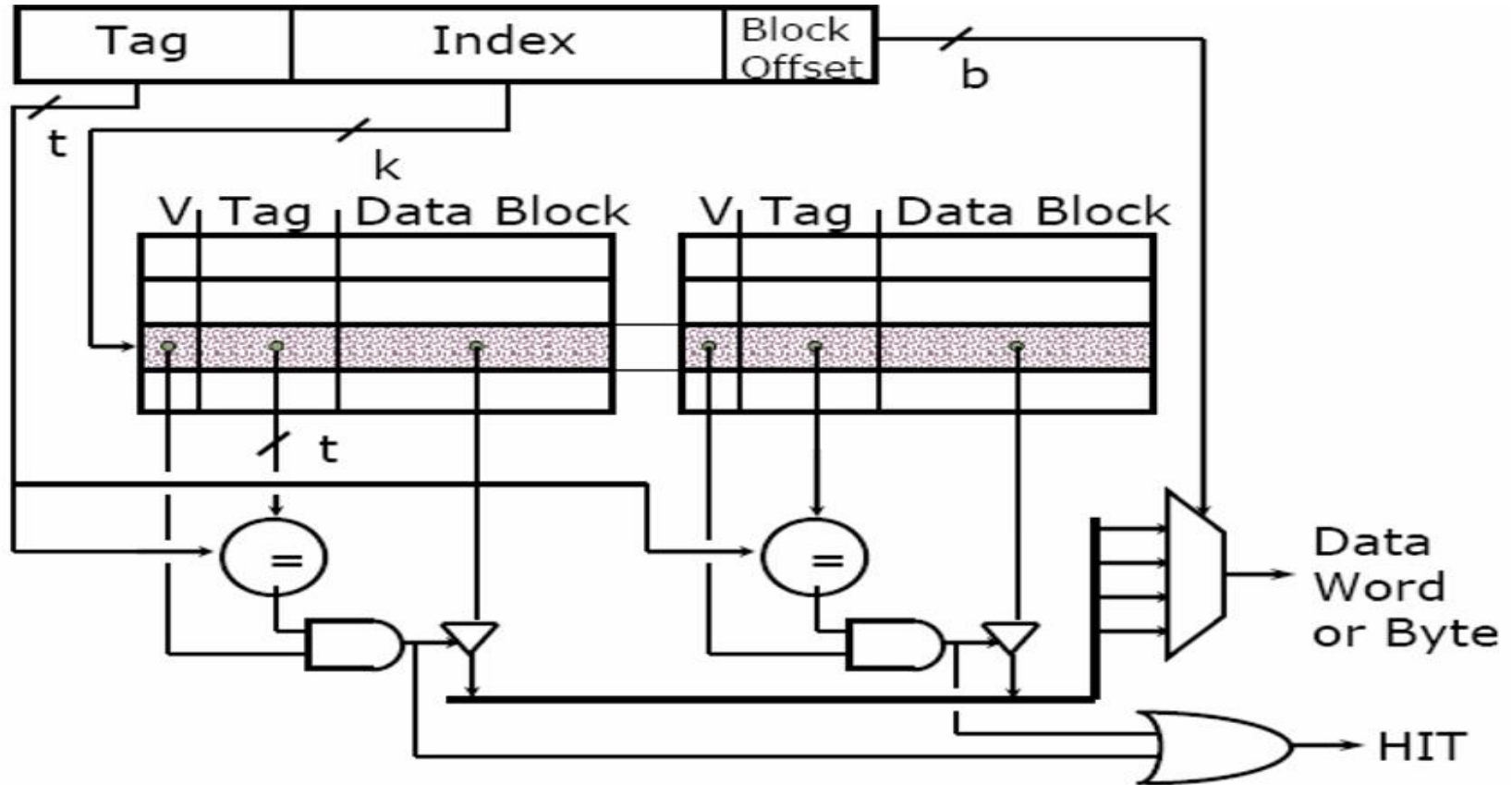


Accessing Set Associative Caches

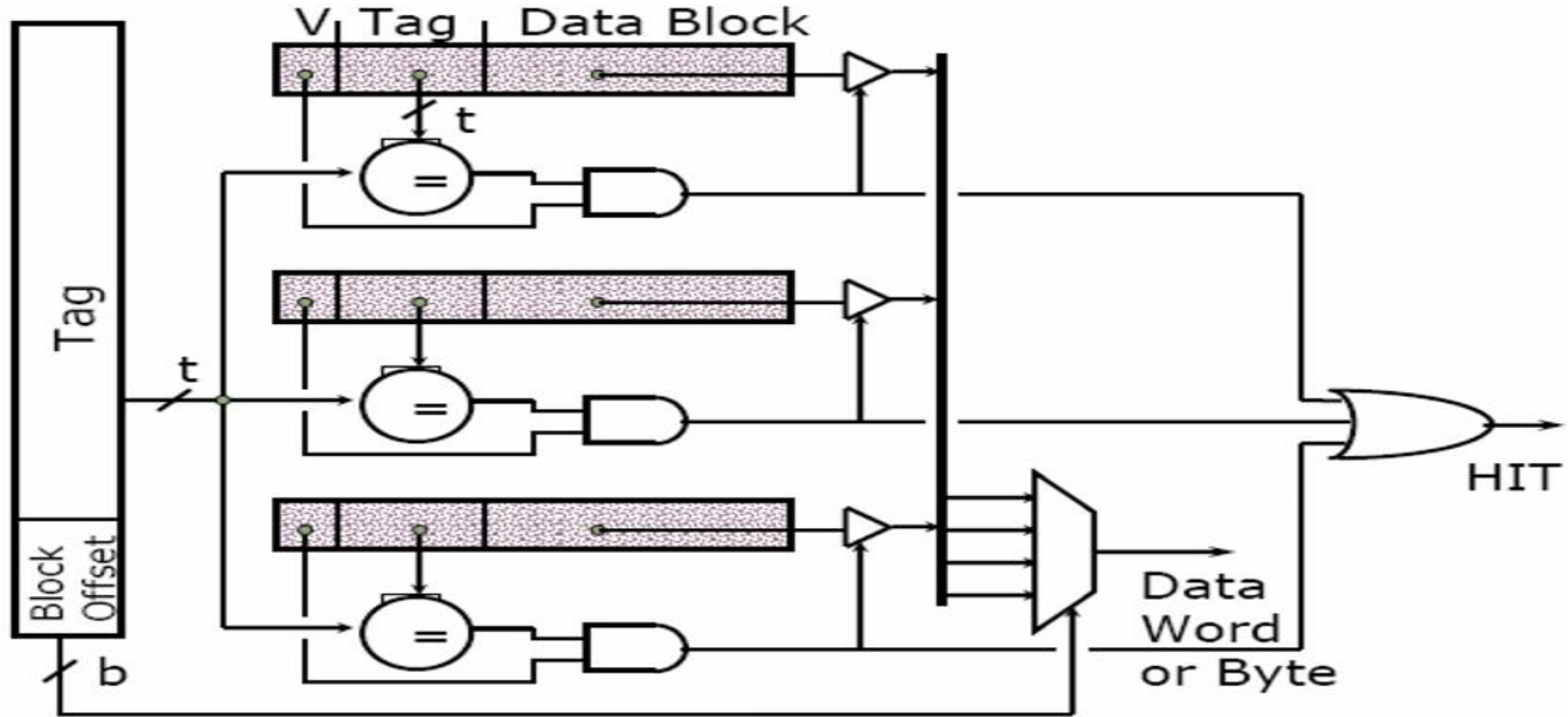
- ❖ Word selection is done same as direct mapped cache but chosen only on the line has produced a hit.



Block Identification – Set Associative



Block Identification – Fully Associative



Index and Offset Calculations

A cache has 512 KB capacity, 4B word, 64B block size and 8-way set associative. The system is using 32 bit address. Given the address 0X ABC89984, which set of cache will be searched and specify which word of the selected cache block will be forwarded if it is a hit in cache?

$$\# \text{ sets} = \text{CS}/(\text{BS} \times \text{A}) = 2^{19}/(2^6 \times 2^3) = 2^{10} = 1024 \text{ sets}$$

1 word = 4B , Hence 64 byte block has 16 words

Tag = 16	Index = 10	Offset = 6 (4+2)
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0x ABC89984 = 1001 1001 1000 0100 → Set 614, word 1

0x 485669AC = 0110 1001 1010 1100 → Set 422, word 11

Mapping Calculations

The address of a word in a byte addressable 1MB physical memory is 0xAB8F2. This word upon bringing to the cache is mapped to set 30. The word length of the processor is 16 bits. How many words can be accommodated in each cache block?

- ❖ 0xAB8F2 → 10101011100011110010
- ❖ 30 → 11110: We can see the set index 30 (11110) above. RHS of set index is byte offset.
- ❖ offset → 010 → 8 bytes. one word is 2 bytes [16 bits] therefore 8 byte offset can accommodate 4 words



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